Product Return Request — AWS Deployment Test Cases

MIS 3371 | Section 26252 | Group 3

Deployed frontend URL	https://staging.d1sg7ta5nv9m14.amplifyapp.com
API Gateway endpoint	https://iqpj8rd7ff.execute-api.us-east-1.amazonaws.com/api/returns (POST)
Lambda function	MIS3371-ReturnRequest (Node.js, execution role: LabRole)
Test dates assume	Today's date = 2026-11-07. If testing on a different day, shift every Purchase Date by the same number of days. The backend computes “today” using the America/Chicago (Houston) time zone, so date boundaries match the local calendar day regardless of server time.

Rules under test

BR-1 a return must be requested within 30 days of the purchase date (day 30 is allowed, day 31 is not). **BR-2** items marked “Was this a Final Sale or Clearance item?” = Yes cannot be returned. **BR-3** the quantity returned cannot exceed the original quantity purchased. **BR-4** a defective item valued over $200 requires supervisor approval (exactly $200 does not). **Input validation** (in Lambda): required fields, a real calendar date that is not in the future, whole-number quantities of at least 1, non-negative item value, allowed reason / resolution / return method / final-sale value (Yes or No), and a replacement SKU when Replacement is chosen.

Part 1 — Test through the deployed frontend

Open the frontend URL and enter the values below, then press Submit. Unless a row says otherwise, use Customer ID **C-1001**, Order Number **ORD-5001**, SKU **SKU-1001**, Final Sale/Clearance **No**, Preferred Resolution **Refund**, Return Method **Mail**. Every submission is sent to Lambda through API Gateway; the result panel under the Submit button shows the backend's response (a heading, then the message).

#	Type	Final Sale?	Qty Ret. / Orig.	Purchase Date	Reason	Item ($)	Resolu-tion	Expected result (heading: message)
1	Valid	No	1 / 2	2026-10-28	No Longer Needed	50	Refund	**Approved:** Approved.
2	Valid	No	1 / 1	2026-10-28	Wrong Item	80	Replace ment, Repl. SKU SKU-550 0	**Approved:** Approved.
3	Business rule (BR-4)	No	1 / 1	2026-10-21	Defective	250	Refund	**Approved — pending supervisor review:** Approved, pending supervisor review (defective item over $200).
4	Invalid (BR-1)	No	1 / 1	2026-09-23	Wrong Item	40	Refund	**Not approved:** Return request is outside the 30-day return window.
5	Invalid (BR-2)	**Yes**	1 / 1	2026-10-28	No Longer Needed	50	Refund	**Not approved:** Final-sale and clearance items are not eligible for return.
6	Invalid (BR-2)	**Yes**	1 / 1	2026-10-28	Defective	15	Refund	**Not approved:** Final-sale and clearance items are not eligible for return. (BR-2 blocks the item before BR-4 is even checked.)
7	Invalid (BR-3)	No	3 / 2	2026-10-30	Other	60	Refund	**Not approved:** Return quantity exceeds the original quantity purchased.
8	Invalid input	No	1 / 2	2026-11-08 (future)	No Longer Needed	50	Refund	**Not approved:** Purchase date cannot be in the future.

#	Type	Final Sale?	Qty Ret. / Orig.	Purchase Date	Reason	Item ($)	Resolu-tion	Expected result (heading: message)
9	Boundary (30 days)	No	1 / 1	2026-10-08	Other	75	Refund	**Approved:** Approved. (Exactly 30 days is still inside the window.)
10	Boundary (31 days)	No	1 / 1	2026-10-07	Other	75	Refund	**Not approved:** Return request is outside the 30-day return window.
11	Boundary ($200)	No	1 / 1	2026-10-28	Defective	200.00	Refund	**Approved:** Approved. (Exactly $200 does not need supervisor review.)
12	Boundary ($200.01)	No	1 / 1	2026-10-28	Defective	200.01	Refund	**Approved — pending supervisor review:** Approved, pending supervisor review (defective item over $200).
13	Boundary (quantity)	No	2 / 2	2026-10-28	No Longer Needed	50	Refund	**Approved:** Approved. (Returning the full original quantity is allowed.)

Part 2 — Backend validation through the API

The browser's built-in checks stop empty required fields and quantities below 1 before they are sent, so these cases are sent straight to the API (curl, Postman, or the Lambda console test tab). Send the base body below with only the listed change. A rejected request returns HTTP 400; an approved request returns HTTP 200.

```
curl -X POST https://iqpj8rd7ff.execute-api.us-east-1.amazonaws.com/api/returns -H "Content-Type: application/json" -d 'BODY'
```

```
Base body: {"customerId":"C-1001","orderNumber":"ORD-5001","sku":"SKU-1001","purchaseDate":"2026-10-28","originalQuantity":2,"quantity":1,"itemTotal":50,"reasonCode":"No Longer Needed","preferredResolution":"Refund","returnMethod":"Mail","finalSale":"No","replacementSku":""}
```

#	Scenario	Change to the base body	HTTP	Expected message
A1	Quantity of 0	`"quantity":0`	400	Quantity to return must be a whole number of at least 1.
A2	Replacement with no replacement SKU	`"preferredResolution":"Replacement"` (leave `replacementSku` empty)	400	Replacement SKU is required when requesting a replacement.
A3	Missing Customer ID	`"customerId":""`	400	Customer ID is required.
A4	Unknown reason code	`"reasonCode":"Changed my mind"`	400	Reason for return must be one of: Defective, Wrong Item, No Longer Needed, Other.
A5	Negative item value	`"itemTotal":-5`	400	Item value must be a number of 0 or more.
A6	Purchase date is not a real date	`"purchaseDate":"2026-02-31"`	400	Purchase date must be a valid date (YYYY-MM-DD).
A7	Missing/blank Final Sale value	`"finalSale":""`	400	Final sale / clearance must be Yes or No.
A8	Body is not valid JSON	`send {oops as the body`	400	Request body must be valid JSON.

Notes

Expected outcomes match the backend test suite (backend/returnRules.test.js, 20 of 20 passing) and the Lambda test events in the backend folder. All business rules and input validation are enforced by the backend; the frontend only shows live hints, so every result in Part 1 comes from Lambda through API Gateway. Every row in both tables above was re-verified directly against the live, deployed API on 2026-11-07.